PA consistency machines: cross-archive constructions and the sub-196 search

Research prepared for Vladimir Reshetnikov

13 November 2026

Abstract

The requested binary Turing machine with fewer than 196 working states has not been constructed in this investigation. This report states that limitation explicitly. It compares all 27 incoming archives at ProveIt commit `906367250a7f96c923147b10df392294bced7be5`, including nested historical inputs, and distinguishes 20 designated 196-state transition graphs from rejected smaller experiments.

The positive result is a new composition of improvements from different recent reports. The diagonal-vacuity axiom emitter and its separate-head variant combine with the shorter cached-head Shift and MP routines. Two complete 196-state tables have respectively 3,676 and 3,649 source instructions; both reach the exact clean seed after 57,922 binary steps. Combining each with the zero-jump optimization gives two further 196-state tables. The archive contains complete transition tables, source grammars, symbolic runtime checks, independent source and startup checks, a full comparison inventory, and further bounded reduction experiments. These are implementations and mathematical evidence at explicitly stated proof boundaries, not a new end-to-end Lean verification of the internal consistency biconditional. No negative search result here is a general lower bound of 196.

Contents

1 The requested result and the actual result **3**

2 Comparison of all supplied designs **3**
- 2.1 Coverage and deduplication . . . 3
- 2.2 The major lineages . . . 4
- 2.3 The twenty selected 196-state graphs . . . 4
- 2.4 What fails in the supplied 195-state files . . . 5

3 The new cross-archive constructions **5**
- 3.1 Why these changes can be composed . . . 5
- 3.2 The axiom changes . . . 6
- 3.3 The shorter command exterior . . . 6
- 3.4 The four complete 196-state outputs . . . 7

4 The zero-jump composition **7**

5 What the checks establish **8**
5.1 Literal startup and complete tape equality 8
5.2 Uniform primitive contracts 8
5.3 From positive instruction blocks to halting equivalence 9

6 The consistency calibration and its limits **9**

7 New reduction searches **10**
7.1 Joint allocation across recent reports 10
7.2 Direct negation with the diagonal calculus 10
7.3 Whole-instruction endpoint searches 10

8 What transfers from ZFC and iPRA **11**
8.1 An automatically fresh gate parameter 12

9 Further research questions **12**

10 Reproduction and artifact guide **13**

A Catalogue of all 27 incoming archives **15**

B A command compiler for the two cross-grafted emitters **17**

C Complete table of the 3,649-site construction **20**

1 The requested result and the actual result

The target is a concrete machine K such that

$$\#\text{WorkingStates}(K) \leq 195, \qquad \text{PA} \vdash (\text{Nonhalt}(K) \leftrightarrow \text{Con}(\text{PA})).$$

The machine convention is inherited from the supplied constructions: one two-way infinite tape, alphabet $\{0, 1\}$, all-zero initial tape, and exactly one-cell left or right moves. A distinguished halting label is excluded from the working-state count. No alphabet enlargement, preloaded program tape, stationary move, or change in counting convention is used to manufacture a reduction.

The strict state-count target remains unmet. The new constructions in this packet all have 196 working states, except for a further direct-negation composition with 197. The four 196-state tables are useful intermediates because they combine previously separate logical and command-level improvements and are accompanied by complete checks. They must not be reported as 195-state successes.

The current comparison is pinned to the commit above. The imported James Chang PA196 and PA209 snapshots are respectively `407cdc356f6b586635e71ac92be394255f4bfea4` and `69d98241a721ead20b4f0b17df9062950b40fbfe`. The reports supplied inside incoming archives can cite earlier ProveIt revisions; their exact bytes and ancestry are retained in the inventory. Thus an older report's assertion about the number of incoming archives is not silently treated as a statement about this newer snapshot.

2 Comparison of all supplied designs

2.1 Coverage and deduplication

The audit opens every top-level incoming ZIP, recursively follows nested ZIPs, and hashes archive payloads before revisiting them. It parses literal transition files without executing programs merely because they appear in an archive. The fresh counts are:

Object counted	Count
Top-level incoming ZIP archives	27
Distinct archive payloads, including nested inputs	40
Maximum nesting depth	7
Literal `.tm` members in distinct payloads	1,466
Byte-distinct transition files	521
Distinct parsed transition tables	515
Unparsable transition files	0
Selected designated 196-state tables	20
Rooted action-preserving graph classes among those tables	20
Occurrences of 195-state diagnostic files	14
Byte-distinct 195-state forms	8
Identified rejected 195-state constructions	6

The broad counts include repeated references, ancestors, experimental tables, and small synthetic controls. They are not counts of valid consistency machines. In particular, the presence of a 195-state file does not establish its intended simulation.

The complete per-archive comparison is `comparison/current_archive_comparison.csv`; the full member-level inventory is `comparison/current_inventory.json`. Exact archive members, table

digests, source bindings, and evidence descriptions for the 20 selected machines are in `comparison/selected196.json`. The corresponding tables and independent checker are included. These files make the scope "all archives" auditable without copying hundreds of megabytes of recursively repeated inputs into this packet.

2.2 The major lineages

Family	States	Construction and comparison significance
James Chang PA209	209	Direct predecessor of the published PA196; 3,948 source instruction occurrences.
Original U208	208	A distinct U209 lineage with 3,909 source instructions and an entry-history fold. Nine original presentations collapse to two rooted graph classes.
U208 helper layouts	208	Eighteen further helper choices give eighteen more classes; the designated corpus has 27 presentations and 20 classes.
Published PA196	196	3,812 source instructions; transposed quantifier coding, six main axiom heads, a guarded gate, and complementary reader sharing.
Later PA196 layouts	196	Reassociations, register choices, cached-head commands, scanner repairs, diagonal-vacuity emitters, and a zero-jump specialization. Their proof scopes differ.
Direct negation	197	Shortens arithmetic source blocks, but its reader sharing is less favorable. Source size alone does not predict physical state count.

U208's selected U209 parent must not be identified with James Chang's PA209 merely because both parents have 209 working states. The literal tables and source identities distinguish them. Likewise, dense renumbering, renamed files, and copies in nested archives are not new constructions.

The PA196 paper allocates 36 states to the engine, 13 to leaf routing, and 147 to readers representing 159 distinct internal source-tree nodes [1]. Those counts already include the implemented sharing. Subtracting the reported reader folds from 196 a second time would be an accounting error.

2.3 The twenty selected 196-state graphs

The earlier fifteen-table census consists of the published machine, an A/D register swap, two experimental MP variants, four closing-decrement choices, a shorter-path layout, two rotation layouts, an eight-zero scanner repair, a rebracketing, an H-only Shift layout, and a later twelve-rotation layout. The current snapshot adds the following five different graphs, whose supplied positive checks were freshly replayed:

Added table	Source sites	Instruction cases	Startup steps
Combined196	3,788	4,928	57,922
Diagonal196	3,700	4,815	58,154
Separate196	3,673	4,780	58,154
Scanner-oriented Separate196	3,673	4,780	58,294
Zerojump196	3,812	4,959	58,402

The independent graph comparison starts at the designated initial control, explores the read-0 edge before read-1, and records the write symbol, move direction, and successor class. This is rooted graph isomorphism preserving every action. All twenty graphs differ. Every selected table also has 196 ordinary action-bisimulation classes. Consequently none has an overlooked ordinary transition quotient. This does not exclude a context-dependent or multi-step simulation reduction.

2.4 What fails in the supplied 195-state files

The six rejected constructions have different failure modes:

1. The two-rotation quotient is present in native and dense numbering; its recorded primitive check has 93 failures.

2. `warm_physical_61_8047` has 714 recorded failures. An intended source frame at site 1111 reaches an exact four-transition cycle after 26 transitions.

3. `extended_physical_61_12085` has 777 recorded failures. This audit identifies its table and original receipt without claiming a freshly reproduced additional concrete witness.

4. The seven-zero scanner regrouping returns from source site 1003 to the wrong clean frame. Extending the scanner repairs the implementation and restores the count to 196.

5. The newer short-scanner deletion has 96 unrestricted branch failures and also fails its stronger invariant-conditioned seed check: the first root return after 118 transitions differs from the intended successor in 43 tape cells.

6. The zero-jump/helper fold takes the wrong source successor and increments the wrong register at source site 26. The newest verifier reproduces this on the candidate's blank execution.

A failed proposed simulation is not, by itself, a theorem that the edited table could never have the desired blank-tape property under some different interpretation. The justified conclusion is that none of these files supplies an accepted sub-196 construction with the stated proof.

3 The new cross-archive constructions

3.1 Why these changes can be composed

The newest diagonal emitters change the axiom command, retaining the original 169-site exterior. Combined196 changes the exterior to 145 sites, retaining the original 3,643-site axiom body. These are separate interfaces. The natural new experiment is to bind each revised axiom body to the revised exterior and check every resulting control-flow edge, rather than infer correctness by adding reported savings.

The two grafts use the exact checked diagonal and separate-head grammars. Their AX bodies have 3,531 and 3,504 primitive sites. Thus

$$3,531 + 145 = 3,676, \qquad 3,504 + 145 = 3,649.$$

Their physical allocation retains 134 fixed controls, adds 68 AX reader controls, and performs six simultaneous compatible folds:

$$134 + 68 - 6 = 196.$$

The shorter expanded source does not remove a reader in this allocation.

In the original diagonal layout the AX prefix is the bit word `10101001`. In the Combined exterior it becomes `100101001`. Every AX leaf is transplanted by replacing the former prefix by the latter. The remaining 145 leaves come exactly from Combined196. The independent verifier reconstructs the ordinal graph by prefix intervals, using neither the construction script nor its symbolic executor. It checks every exterior instruction and every incoming and outgoing AX edge under the ordinal shift.

The command return site 3776 in Combined196 becomes site 3664 in the diagonal graft and site 3637 in the separate-head graft. This check matters: a table can be total and have an apparently correct AX block yet return to the wrong command dispatcher.

3.2 The axiom changes

The numeric constructors are based on

$$p(a,b) = \binom{a+b+1}{2} + b, \quad \mathsf{I}(A,B) = 4p(A,B), \quad \mathsf{M}(a,b) = 4p(a,b)+1,$$

$$\mathsf{U}(a,B) = 4p(B,a)+2, \qquad \mathsf{E}(a,b) = 4p(a,b)+3.$$

Code 0 denotes truth, code 1 denotes the false target, and $\mathsf{N}(B) = \mathsf{I}(B,1)$. The quantifier's two pairing arguments are transposed. Every variable index in the formula decoded from a positive code B is strictly less than B; code 0 has no free variable. Hence

$$\Gamma(B) = \mathsf{I}(B, \mathsf{U}(B,B))$$

is a valid diagonal-vacuity instance: $B \to \forall x_B B$.

The diagonal construction replaces the equality-vacuity component of the old guarded gate by $\Gamma(c)$. Writing V for the unchanged membership-vacuity clause and Q for the unchanged quantified separation clause, its top formula is

$$F_\Gamma(a,b,c) = \mathsf{N}\big(\mathsf{I}(\Gamma(c), \mathsf{I}(V,Q))\big).$$

The old guard $c \neq a, b$ remains. The separate-head construction emits $\Gamma(c)$ as an additional stack item and leaves

$$G(a,b,c) = \mathsf{N}(\mathsf{I}(V,Q))$$

as its guarded top. All six principal heads, the restricted propositional head, and the generalization item retain their established roles.

These two axiom presentations were proposed in the supplied reports; their logical simplification is not claimed as a discovery here. The new result here is their composition with the newer exterior, its exact physical realization, and its independent binding checks. The prior diagonal derivation recovers the reverse membership rewrites, neighborhood equality, and equality witness interfaces used by the PA refutation compiler [4]. It explicitly retains membership vacuity; discarding that clause as well would leave a circular proof.

The free-algebra emitter check compares complete constructor trees, not a sample of small numeric inputs. The diagonal graft has 191 macro items and nine pushed stack entries, including the saved top. The separate-head graft has 190 macro items and ten pushed entries. Its new head is an implication and therefore is not the literal false code 1.

3.3 The shorter command exterior

The Combined exterior changes two commands. Shift clears the old top T, copies the cached head H into T, and tests the old value of H against 1. MP's successful branch stores its consequent in H, with $T = D = 0$, and does not push an unnecessary zero. Its failed branch retains the appropriate offset in D and has $T = H = 0$.

These changes require a corresponding proof-command compiler; they are not simple equality of the old and new commands on every store. The supplied Combined report gives the arithmetic loop invariants and a compiler adapted to these effects [3]. Our new source binding preserves those entire exterior fragments.

For example, the Shift pre-guard transformation $(T,H,X) = (t,h,0) \mapsto (h,h,0)$ takes $2t+7h+3$ source instructions: the clearing loop costs $2t+1$, the copying loop costs $4h+1$, and the restoration loop costs $3h+1$. The successful MP transfer $(D,H) = (y,0) \mapsto (0,y)$ costs $3y+1$ before its return

jump. Induction on t, h, y establishes these formulas for arbitrary naturals. They are not estimates based on bounded execution.

The nine-entry and ten-entry stack conventions must be distinguished when composing the command compiler. The extra diagonal head shifts the selectors for the old heads by one. The packet's independent review includes explicit algebra for the two stack lengths. It does not assume that an old nine-entry selector works unchanged for ten entries.

3.4 The four complete 196-state outputs

Construction	Source sites	Cases	Startup
Diagonal AX + Combined exterior	3,676	4,784	57,922
Separate diagonal head + Combined exterior	3,649	4,749	57,922
Diagonal + Combined + zero jump	3,676	4,785	58,168
Separate + Combined + zero jump	3,649	4,750	58,168

Each table has exactly 392 rows, working states $0, \ldots, 195$, start state 0, and terminal label 196. Dense and native versions, their bijective label maps, source sites, grammars, and receipts are in `machines/`. Appendix C prints the entire table of the 3,649-site construction without the zero-jump specialization.

The first two table SHA-256 digests are respectively

```
75799d7c7c4f27a02d5058a0424bce3ad5c19746022663b1668675914fb89ead
66e34596fa929590cf768ca90856824813af837a0a60fd13e2f1e957cd4f59b6.
```

The zero-jump versions have digests

```
9a7c6280c3aaf0ee9fdf205f7fb27f52a2b79c0c0640def20cafd66d9ee2fffb
4d8c0dfb554698d3b7f400009fedc739af6e5a3b0d5525567165ac7f56ec31aa.
```

An independent novelty check compares the four outputs with all 268 prior literal 196-state table occurrences, representing 127 parsed-distinct tables, including experimental and rejected tables. After a necessary action-signature filter it performs 23,324 rooted canonicalizations, allowing every working label as the prior table's initial control. No prior graph matches any output. The four outputs are also pairwise distinct, and each has 196 graph-reachable controls and 196 ordinary action-bisimulation classes. This establishes novelty within the supplied table corpus, without asserting a global literature priority claim.

4 The zero-jump composition

The newest zero-jump report replaces the forward jump at source site 26 by a decrement of register 6, which is zero at that site on the reachable source run [5]. In native table labels its physical change is

$$(27, 1) \mapsto (0, L, 40) \quad \text{replaced by} \quad (27, 1) \mapsto (1, R, 21).$$

It does not delete a state. Applied to each new graft, it again has no state-count saving, but it changes the available local routing and so is a useful additional search seed.

Lemma 4.1 (Zero-decrement replacement)**.** *Suppose an inductive source invariant says register r is zero whenever control is at site i. Replacing an unconditional jump at i by a decrement of r with the same zero successor preserves the complete source execution from an invariant-satisfying seed, provided all other instructions are unchanged.*

Proof. At i, decrement takes its zero branch and changes no register. Consequently the old and new next configurations coincide. At other sites the instruction is identical. Induction on the source-step count gives equality of the executions and therefore equality of halting. □

The invariant used here tracks a nine-bit positivity mask m: bit r is set precisely when register r is positive. Increment sets its bit; a zero decrement follows the zero branch without change; a positive decrement follows the positive branch with either a positive or zero remainder. This nondeterministic remainder is an overapproximation of all natural values, not an assumption that a positive register equals 1.

The exact clean seed is source site 25 with mask 1, corresponding to register vector $(2, 0, 0, 0, 0, 0, 0, 0, 0)$. The two new finite closures contain 265,964 and 263,244 site-mask pairs. Each contains the seed, is closed under every abstract successor, and has only masks with bit 6 clear at site 26. Concrete source induction therefore proves the required zero condition. Because both complete graphs are checked, this argument does not assume that an invariant computed before the AX change remains valid afterward.

The changed decrement introduces a further abstract positive branch into the source specification, explaining the one-case increase in each uniform physical runtime check. That branch is checked physically too; the source invariant excludes it only in the equivalence with the original jump program.

5 What the checks establish

5.1 Literal startup and complete tape equality

The startup checks execute the literal binary table from its designated state on a blank tape. At the accepted boundary the head is at position -2, the native control is the source root 84, and the physical register list is

$$[0, 0, 0, 0, 0, 0, 2, 0, 0] +\!\!+ [1]^{26}.$$

The check includes the origin marker, complete source path, padding, file boundary, every unary register block, and all nonzero tape cells. A matching prefix or matching logical registers alone would not suffice. The affine gap starts at 48 and can grow in the runtime representation.

The reported startup durations are exact finite facts about these tables. They do not prove nonhalting by themselves. Their role is to connect the blank initial configuration to the hypothesis of the uniform source simulation.

5.2 Uniform primitive contracts

The symbolic executor represents the whole register file and an affine unbounded gap. Every source instruction is checked; every decrement is checked in both branches. Unary register operations are summarized only at the unchanged engine interfaces from the upstream runtime. The fixed engine rows are retained, so those inherited lemmas still apply. Symbolic execution verifies the exact successor frame, including the file effect, rather than merely whether it returns to a plausible control.

The counts 4,784 and 4,749 are source-outcome cases, not a collection of bounded natural-number test inputs. Conversely, their symbolic step counts are not literal binary running times. The independent review binds these source cases to the actual grammar and its resolved control graph, while a separate check binds dense labels to the native table.

These executable checks are mathematical evidence using an inherited runtime interface. They are not advertised as a new Lean kernel build. A hostile mutation test, a long blank prefix, or an implementation's own assertions alone would be a weaker form of evidence.

5.3 From positive instruction blocks to halting equivalence

Lemma 5.1 (Positive-block transfer)**.** *Suppose a deterministic source machine and a deterministic binary machine have related initial configurations after a finite startup. Suppose every nonhalting source step from a related frame has a finite binary simulation of positive length, with no intermediate halt, ending at the next related frame. Suppose source halt corresponds to binary halt. Then the two executions halt together.*

Proof. A finite source halt transfers by concatenating its finite simulation blocks. Conversely, if the binary machine halts after n transitions following startup, a nonhalting source prefix of $n+1$ steps would transfer to a nonhalting binary prefix of at least $n+1$ transitions. Determinism contradicts the alleged earlier halt. Thus a source halt must occur. □

For the internal-PA version, the configuration relations, finite traces, and block constructors must be arithmetized, and their correctness and positive lengths must be proved in PA. The argument then uses finite arithmetic induction; it does not require a truth predicate for arbitrary formulas. This is the useful distinction between a transferable mathematical proof and a completed exported formal theorem.

6 The consistency calibration and its limits

There are three different questions: whether the table is a correct implementation of the displayed source, whether source halting is equivalent to a refutation in the selected numeric calculus, and whether that refutation condition is PA-provably equivalent to a refutation of ordinary PA. State counting answers none of the latter two questions.

The inspected imported Lean development exports

$$\mathrm{PA} \vdash \mathrm{Nonhalt}(M_{196}) \to \mathrm{Con}(\mathrm{PA}),$$

and an external Lean-metatheoretic nonhalting theorem [2]. Its theorem inventory does not export the reverse internal implication. That observation does not prove the converse false. Several incoming reports provide ordinary mathematical arguments for it, and the Combined report supplies especially explicit command and positive-block arguments. The current packet preserves that distinction.

The new grafts reuse two written calibration components: the diagonal calculus's refutation translations, and the Combined exterior's command compiler. The diagonal variant keeps nine stack entries; the separate variant uses the adjusted ten-entry selectors. The independent binding proof checks that these are the fragments actually executed. Subject to those inherited uniform arithmetic proof translations, Lemma 5.1 and the exact startup give the same PA consistency calibration for the new tables. This packet adds no new closed Lean theorem asserting that entire biconditional.

The reverse arithmetic direction must be syntactic. Interpreting membership by a binary-bit relation and saying that every derivable formula is true in the standard model is not, by itself, an internal PA proof of $\mathrm{Halt} \to \neg\,\mathrm{Con}(\mathrm{PA})$. A sufficient internal construction is: extract a coded numeric-calculus refutation from a halted finite trace; translate each axiom and rule by uniformly generated finite PA proofs; and concatenate them into a PA refutation. Quantifier recoding, freshness, the code-0 alias, and the zero-alias stack convention $p(0,0) = 0$ must all be handled by that same code-level argument.

The forward refutation-to-halting direction likewise needs an executable proof-command compiler and exhaustive enumeration. Changing the number or order of pushed heads affects that compiler even when each emitted formula is logically harmless. These are the reasons the artifact keeps source binding, emitter semantics, runtime contracts, and calibration evidence as separately identified components.

7 New reduction searches

7.1 Joint allocation across recent reports

The new composition screen reconstructs the two primary grammars and 98 saved cost-tied grammars under the Combined fixed control. All 100 retain the same nominal physical total of 196. An additional joint reader screen includes the scanner-oriented grammar, for 101 inputs. It allows compatible reader folds involving the fixed exterior as well as the AX grammar, and separately checks whether an AX reader can be replaced by an existing fixed control on the admitted local contexts. No extra saving results. Repeating that 101-input screen after the zero-jump change also gives no sub-196 table.

These are finite, exactly retained inputs, not a claim to enumerate all grammars. The primary new tables receive full source/runtime/startup checks; the other tied layouts are search inputs, not 98 additional claimed correct machines. The saved table and source digests permit independent replay of the positive outputs without repeating a heuristic trajectory.

7.2 Direct negation with the diagonal calculus

The new direct-negation composition replaces eight occurrences of the push-top, increment, cons sequence by an existing arithmetic negation routine, under its scratch-zero contract. The retained best diagonal layout has 3,468 source sites and 197 working states; all 4,503 physical instruction cases pass. Grafting that AX onto the Combined exterior reduces the source to 3,444 sites with 4,472 passing cases and exact 57,922-step startup, but it still uses 197 states.

Concretely, with $S = R = 0$ and $T = t$, increment R twice and execute the old `node64`. Its triangular accumulation followed by fourfold transfer leaves

$$T = 4\big(2 + \mathrm{tri}(t) + t\big) = 4\big(\mathrm{tri}(t+1) + 1\big) = \mathsf{I}(t, 1), \qquad \mathrm{tri}(t) = t(t+1)/2.$$

It restores $S = R = 0$ and preserves the other registers. Each replacement saves 29 primitive sites, giving 232 in total. Clean scratch is the inherited macro interface transported through this composition argument; the free-constructor check does not track these scratch registers, and no new formally checked AX plan is asserted. The separate theory packet in `research/theory_study/` records this distinction and the complete 197-state reproduction.

The retained allocation has 68 AX reader nodes and five compatible folds, yielding $134+68-5 = 197$. This is a concrete illustration of the tradeoff: a substantial source reduction loses a sharing opportunity. The bounded search evaluates 100 initial composed grammars and 38,400 seeded mutation slots. Its negative outcome is restricted to that experiment. The scratch-zero arithmetic obligation is preserved as an explicit hypothesis; a uniform physical instruction check does not itself prove that higher-level macro precondition.

7.3 Whole-instruction endpoint searches

The further physical search changes the equivalence boundary. Earlier screens primarily require one parent binary step to match a short child block. Here a candidate may redistribute work over a complete source instruction before reaching the required frame. This admits a different family of potential reductions.

The baseline fixture contains 4,959 complete source-outcome frames. All primary registers are zero except that the tested positive decrement register is one. These are concrete filters, unlike the unbounded symbolic contracts used for the positive constructions. All baseline endpoints are checked literally; 59 selected cases cover all 392 rows. A guarded fixture uses the 274,880-pair source positivity invariant and has 4,956 representative feasible-branch cases. It still covers every baseline row. These representative concrete frames are not asserted to occur in the actual blank run.

For each deleted state, a keeper is selected, all incoming edges are redirected, and either row of the keeper may be replaced by any legal action. The full 196-state screen examines

$$196 \cdot 195 \cdot 2 \cdot 784 = 59,928,960$$

row assignments and finds no survivor. The guarded exact version and a relaxed-frame version each examine the 49-state engine/router subset, with 14,982,240 assignments per version, and find none. The relaxed version allows a different gap of at least 48 and arbitrary valid spare blocks while retaining the source path, head, marker and all nine primary register values.

A separate helper extension permits the changed row to belong to any retained control. Its completed prefix consists of deleted states `0,1,2,3,4,5,6,7,9,10`; it checks 29,495,648 assignments with no survivor. The interrupted work on state 14 is excluded from that count. Failure-trace intersections restrict which single helper rows can possibly repair all failed tests within the same budget.

Each concrete test allows at most $\max(800, 2s + 64)$ candidate transitions, where s is the parent instruction duration, and uses a 256-cell tape array. Unexpected halt, leaving the array, or failing to reach the required endpoint within the budget rejects the assignment. A wrong intermediate root visit alone does not reject it. An independent set-of-cells simulator checks all 4,959 baseline endpoints and all 4,959 endpoints after a genuine two-step head round trip is inserted before root/read-0. Deliberate halt and infinite-motion controls are rejected for the appropriate reasons.

The search counts are assignments and overlap between lanes. A rejected assignment may merely require a longer simulation or a different representation. Thus the conclusion is precisely that no tested assignment passes the stated finite endpoint filter. It is not a proof of different blank halting behavior or a general lower bound. Complete sources, fixtures, positive controls, limits and logs are retained under `research/macro_boundary_study/`.

8 What transfers from ZFC and iPRA

The useful transferable objects are compiler and proof techniques: entry-history information stored on tape, shared readers, source reassociation with explicit jump scopes, symbolic register contracts, and replacing a primitive schema by a derivable smaller interface. Those techniques can change the implementation while preserving the selected searched calculus. Their applicability has to be checked at the concrete PA source and table [8].

In particular, the ZFC K215 construction combines a command-local field reader, command ordering, and a partial selector while retaining earlier marker folds. Those are opportunities to reconsider a runtime interface, not numerical state savings transferable independently of that interface. Its compiled external Lean/Rocq endpoint and the separate internal-PA endpoint for the 240-state table have different scopes; the ZFC directory explicitly distinguishes them [8].

The final U208 history fold cannot simply be renamed into PA196. Its parent has a mixed-direction same-target predecessor motif which the PA196 table does not contain. That is a finite obstruction to that particular transplant, not an impossibility result for every history fold. The ZFC construction's theory or a smaller iPRA checker is also not a substitute for the PA refutation search: consistency of those theories is a different calibration problem. A replacement needs explicit PA-provable transformations of refutations in both directions.

The concise-iPRA work is relevant to choosing simple proof formats and keeping a checker small. Checker source length, source instruction count, and binary TM state count are three different objectives. None provides a monotonic bound on either of the others under the present compiler. Its one-sided transport and merged introduction rules illustrate a useful method: first prove that a syntactic restriction preserves derivability, then remove the corresponding checker distinction [9]. For the PA

gate restriction below, freshness is proved but the analogous derivability translation is still missing. Likewise, merely switching to full intuitionistic arithmetic requires a negative-translation/refutation argument and a new machine implementation; it does not yield a state count for free.

8.1 An automatically fresh gate parameter

A further new idea is to make the gate's freshness condition automatic by padding its property with an equivalent formula. Let D, H, A be formula codes, put $B = D + 1$, and define

$$T = \mathsf{I}(B, \mathsf{I}(H, B)), \qquad A' = \mathsf{I}(T, A).$$

The decoded T is an instance of the propositional tautology $B \to (H \to B)$, so A' is propositionally equivalent to A. Because $B > 0$, none of these pairing nodes suffers the $\mathsf{I}(0, 0) = 0$ collapse. Strict component growth for positive constructor codes gives $A' > D, H$. The label A' is also fresh in its own decoded formula. Thus a gate emitted with property code A' automatically satisfies both inequalities against D and H.

The $+1$ is essential. Without it, $D = H = A = 0$ gives $T = A' = 0$, so strict freshness fails. This edge case is a reason to reason about actual numeric constructors, rather than only abstract formula trees.

This lemma does not by itself prove a replacement calculus complete. The old gate supplies membership vacuity at arbitrary fresh labels; the restricted padded gate supplies it at specially generated labels. Recovering the old vacuity and binder-renaming interface requires a new syntactic argument or a revised PA proof compiler. It would be circular to assume unrestricted predicate logic while deriving the axioms that are meant to establish that logic.

In the bundled diagonal variant there is a further obligation: padding before emitting the diagonal clause gives only $\Gamma(A')$ for padded codes, rather than every $\Gamma(B)$. Preserving a separate original $\Gamma(A)$ head could avoid that particular restriction, but would not settle the arbitrary-vacuity interface.

The attempted source implementation pads only after the old heads have been emitted, preserving their arbitrary parameters. Its retained heuristic layout costs 210 controls, well above the target, and the candidate physical runtime check fails at source site 2413. It is not an accepted 210-state consistency machine. The finite run uses 4,200 mutation slots; it supplies a concrete negative implementation result and a useful local mathematical lemma, not a completed theory replacement. The original inequalities cannot simply be deleted.

9 Further research questions

1. Can whole-instruction equivalence, rather than equivalence of individual binary steps, expose an engine reduction? A successful candidate must then be checked over arbitrary register values and a complete tape relation. The macro searches in this packet explore only specified finite templates.

2. Can the new 3,649-site source be allocated by a compiler that jointly chooses the exterior, AX tree, helper controls, and scanner representation? The current $134 + n - f$ accounting fixes a substantial part of the layout in advance.

3. Can a different representation of the source path distinguish the origin marker with fewer controls without making instruction dispatch or blank startup larger? A shortened detector must be correct on intermediate, partially rewritten paths as well as clean addresses.

4. Can a restricted family of automatically fresh gate instances replace the present two inequality guards while retaining an explicit PA refutation compiler? The outstanding logical interface is as important as the emitter's source cost.

5. Can the entire PA calibration be exported as a uniform coded refutation translation, with an arbitrary-internal-time tape/source reflection theorem? This would make future local reductions easier to certify and would settle the proof-status distinction visible in the current theorem inventory.

These questions describe the remaining work; they are not claimed solutions. No lower bound rules out a 195-state machine. The work here shows exactly which combinations and finite search templates were tried, and supplies better-bound intermediate implementations for a subsequent attempt.

10 Reproduction and artifact guide

Run `python3 verify.py` from the extracted packet root, with assertions enabled. It checks file integrity, the twenty-table comparison, the four new tables, their source/runtime/startup contracts, the zero-jump invariants, and the independent graft review. The optional longer search commands and their limits are described in the corresponding research directories. A successful default run does not mean the strict target was achieved: the machine-readable status explicitly records that it was not.

The optional command `python3 research/verify_direct_negation_combined.py` replays the 3,444-site, 197-state composition. From `research/theory_study/`, `python3 -B verify.py` replays the 3,468-site composition, and `python3 -B replay_padded_failure.py` reproduces the explicitly rejected padding allocation. Recomputing the entire original archive inventory additionally requires the original pinned incoming ZIPs; their digests and retrieval locations are supplied. Exact copies of the 127 prior 196-state tables are included, so the default positive replay also checks the complete graph novelty comparison without those original ZIPs.

The packet contains the complete report source and PDF; full literal machine tables and source grammars; the 27-archive comparison and recursive inventory; independent binding and graph checks; bounded-search sources, inputs, and receipts; exact source pins; and retained upstream licenses. The original repository and its imported publication trees were not modified.

The executable checkers use Python 3 and its standard library. Repeating the C++ search screens additionally requires a C++ compiler. Compiling this report uses an ordinary LaTeX installation with the packages named in its preamble. No Lean or Rocq build is claimed by this packet.

References

[1] James Chang, *Nonhalting and PA-independence for a 196-state Turing machine*, PA196 snapshot `407cdc356f6b586635e71ac92be394255f4bfea4`. Imported paper and exact table in ProveIt's `Computability/BusyBeaver/Research/PA/jamesyc-upstream/pa196/`.

[2] ProveIt, `pa196-vs-u208/COMPARISON.md`, `PA-BRIDGE.md`, and the imported PA196 `docs/result.md`, at the pinned commit.
https://github.com/VladimirReshetnikov/ProveIt/tree/906367250a7f96c923147b10df392294bced7be5/Computability/BusyBeaver/Research/PA

[3] Supplied report `docs/incoming/conpa_tm_study_2026-11-13.zip`, especially `chapters/combined_commands.tex` and `chapters/combined_source_macros.tex`. Exact relevant inputs retained under `inputs/combined/`.

[4] Supplied report `docs/incoming/pa_consistency_comparison_and_new196_2026-11-13.zip`, especially `axioms/proof/diagonal_prior.tex`, `axioms/proof/changes.tex`, and the 196-state axiom grammars. Exact relevant inputs retained under `inputs/axioms/`.

[5] Supplied report `docs/incoming/pa_tm_research_2026-11-13 (2).zip`, `verification/source_zero_jumps/` and `verification/zero_jump_machine/`.

[6] Supplied report `docs/incoming/pa_below196_research.zip`. Its earlier census and bounded searches are distinguished from the new 27-archive audit.

[7] Vladimir Reshetnikov, *ProveIt*, pinned input commit and complete incoming directory. https://github.com/VladimirReshetnikov/ProveIt/tree/906367250a7f96c923147b10df392294bced7be5/docs/incoming

[8] ProveIt, *Smaller machines for Con(ZFC)*, directory overview at the pinned input commit. An exact copy is retained as `inputs/related/ZFC_README.md`. https://github.com/VladimirReshetnikov/ProveIt/tree/906367250a7f96c923147b10df392294bced7be5/Computability/BusyBeaver/Research/ZFC

[9] ProveIt, *Concise presentations of intuitionistic primitive recursive arithmetic*, directory overview at the pinned input commit. An exact copy is retained as `inputs/related/ConciseIPRA_README.md`. https://github.com/VladimirReshetnikov/ProveIt/tree/906367250a7f96c923147b10df392294bced7be5/Logic/PrimitiveRecursiveArithmetic/ConciseIPRA

A Catalogue of all 27 incoming archives

This catalogue distinguishes newly proposed tables, repeated inputs, source-only proposals, and rejected implementations. A listed numerical size is not an assertion that the entire proof has been formalized. Full archive digests and member-level provenance are retained in the comparison CSV and JSON files.

Table 1: Archive-by-archive comparison at the pinned ProveIt commit.

Archive	Distinct contribution and validation boundary
`Below_U208_Research_2026-10-29.zip`	Comparison and bounded local, source, and offset searches in the U208 family. No smaller complete machine is supplied.
`ConPA_below208_research_2026-10-29.zip`	Source permutations, two-row screens, and a logical obstruction in the U208 family. No sub-208 construction is established.
`PA_consistency_TM_research_2026-11-13.zip`	Fourteen-table census. A 3,573-site source has 157 grammar nodes, but no new 196- or 195-state physical table is supplied; a residual-MP proposal also remains a source-level experiment.
`U208_ConPA_2026-10-29.zip`	Original U208 class A, using the 172/211 entry-history fold. Other presentations of the same table are deduplicated.
`U208_ConPA_Conditional_Context_2026-10-29.zip`	Original U208 classes A and B, presented through a conditional-history invariant. The proof presentation does not create an additional graph.
`U208_ConPA_Entry_Context_2026-10-29.zip`	Original U208 classes A and B, with an entry-context proof presentation.
`U208_ConPA_History_2026-10-29.zip`	Original U208 classes A and B, with a predecessor-history proof presentation.
`conpa_tm_comparison_and_reduction_search.zip`	Two experimental 196-state MP variants with bounded frame checks. A rejected Shift edit halts after 43,563 blank steps; these proof scopes are distinguished from the later uniform Combined check.
`conpa_tm_comparison_and_search.zip`	An A/D register-swap 196-state table. A rotated 195-state proposal fails validation; an unmodified scanner shortcut halts after 699 blank steps.
`conpa_tm_study_2026-11-13.zip`	Combined196: H-only Shift and MP result in H, 3,788 source sites, 4,928 uniform cases, and 57,922-step startup. A direct-negation variant has 197 states; a repaired 157-node source needs 207 physical states.
`pa196_research_2026-11-13.zip`	A 157-node source without a corresponding accepted binary table, a lost-helper obstruction, and negative bounded two-row synthesis results.
`pa208_research_2026-10-29.zip`	Eighteen additional pairwise nonisomorphic 208-state helper layouts. They extend the two original U208 graph classes to twenty designated classes.
`pa_below196_research.zip`	An earlier 23-archive census with fifteen selected 196-state graphs. Equivalent gate spellings, a restricted implication-matrix minimum, and bounded physical-fold searches do not give a sub-196 machine.
`pa_consistency_comparison_2026_11_13.zip`	Quantifier-head splitting with Hilbert certificates and twelve combined source/table experiments. The best combined negation/MP layout has 197 states; further gate and padding searches do not improve it.
`pa_consistency_comparison_and_new196_2026-11-13.zip`	Three diagonal-calculus 196-state layouts: diagonal, separate head, and scanner-oriented separate head. A hybrid passes frame contracts but its tested starts escape outward. A short-scanner 195-state proposal fails its clean seed.
`pa_consistency_reduction.zip`	Rebracketed196 and H-only Shift196, plus integrated witness-compiler experiments costing 203 or 206 states.

Archive	Distinct contribution and validation boundary
pa_consistency_research.zip	A 158-node source compiles to 196 states after an eight-zero scanner repair. The shorter uncorrected scanner is not an accepted reduction.
pa_consistency_research_2026-10-29.zip	A source reduction from 3,909 to 3,904 sites increases the relevant DAG allocation from 197 to 200 nodes. No smaller literal table is obtained.
pa_consistency_research_2026-11-13 (2).zip	A further distinct 196-state twelve-rotation layout; a 198-state diagonal-vacuity layout; a halt-at-successful-MP command theorem; and reduced joint register abstractions.
pa_consistency_research_2026-11-13.zip	A 199-state diagonal-vacuity implementation and a 197-state negation peephole. Neither satisfies the strict target.
pa_tm_comparison_and_reductions.zip	Two rotated 196-state layouts and an equivalent-gate compiler with 3,773 source sites and 198 physical states.
pa_tm_extended_research_2026-11-13.zip	Four 196-state closing-decrement variants, with their stated constrained symbolic and startup checks.
pa_tm_research_2026-11-13 (2).zip	Zerojump196 uses the source-site-26 zero-register invariant: 4,959 uniform cases and 58,402-step startup. A 195-state fold fails on the actual blank run. Base-8 and disequality-parity proposals have incomplete implementation or startup obligations.
pa_tm_research_2026-11-13.zip	An alternative 196-state layout with maximum source-path length 36, full symbolic instruction checks, and exact startup validation.
pa_unary_power_cover.zip	A unary-cover theory and proof-stream proposal. No complete binary compilation or sub-196 transition table is supplied.
u208_comparison_research_2026-10-29.zip	A 207-state U208-family proposal rejected by a concrete primitive-frame counterexample.
u208_research_2026-10-29.zip	Original U208 class B as the primary table and class A as an alternative; these repeat graph classes in the companion presentations.

B A command compiler for the two cross-grafted emitters

The diagonal cross-graft retains the combined H-only Shift and MP-in-H commands and replaces AX by the diagonal-vacuity emitter. The separate-head cross-graft additionally pushes the diagonal formula as an axiom head. It is necessary to adjust the proof-command words in the latter case. This section gives one argument covering both constructions.

Write a store as $\langle t, w, d, h\rangle$ and write $[a_1, \dots, a_k] ::_C w$ for successive Cantor cons operations, with a_1 at the top. The operations hd, tl are the total Cantor projections, including their zero values. Put

$$I(a, b) = 4p(a, b), \qquad U(a, B) = 4p(B, a) + 2.$$

Shift and MP have the combined-program contracts

$$S\langle t, w, d, h\rangle = \begin{cases} \mathsf{halt}, & h = 1, \\ \langle h, \mathrm{tl}(w), d, \mathrm{hd}(w)\rangle, & h \neq 1, \end{cases}$$

$$\mathsf{MP}_x\langle t, w, d, h\rangle = \begin{cases} \langle 0, w, 0, d + x\rangle, & I(t, d + x) = h, \\ \langle 0, w, d + x, 0\rangle, & I(t, d + x) \neq h. \end{cases}$$

The generic AX interface. Let $m = 8$ for the diagonal cross-graft and $m = 9$ for the separate-head cross-graft. AX pushes m generated formulas and the saved top, so the total pushed-entry counts are respectively nine and ten. Put $D = d + x$ and $H = h + y$. In both cases,

$$\mathsf{AX}_{c,x,y}\langle t, w, d, h\rangle = \langle G_m(H, D, c), [q_1, \dots, q_m, t] ::_C w, 0, 0\rangle.$$

The last two generated entries are

$$q_{m-1} = \rho_8(D, c, H, d), \qquad q_m = U(d, h).$$

For $m = 8$, $q_1, \dots, q_6$ are the unchanged six heads, in reverse emission order. For $m = 9$, $q_1 = \Gamma(c) = I(c, U(c, c))$, and the six old heads occupy positions two through seven. The diagonal formula is pushed unconditionally; a failed numerical guard only replaces the final top G_m by zero. All q_i except q_m have residue zero modulo four, whereas q_m has residue two. The guarded top also has residue zero. Consequently none of these generated values is one.

The independent free-algebra check in `verify_grafts.py` verifies the entire emitted stack and top, the original order of the two operand reads, the unchanged final guard/cleanup macros, and the exact expansion of all 191 or 190 macro items into the candidate source. The updated AX therefore still clears both D and H at command return.

Two total stack identities. For every incoming store, including $t = 1$,

$$\mathsf{AX}_{c,x,y}; S^m : \langle t, w, d, h\rangle \longmapsto \langle \rho_8(D, c, H, d), [t] ::_C w, 0, U(d, h)\rangle,$$

$$\mathsf{AX}_{c,x,y}; S^{m+1} : \langle t, w, d, h\rangle \longmapsto \langle U(d, h), w, 0, t\rangle.$$

The first Shift tests the cleared zero cache. Each subsequent Shift up to the $(m + 1)$st tests one of the generated q_i, and therefore cannot halt. The saved top is not tested until one further Shift. These facts prove the identities directly by the total Cantor head/tail equations. They do not require injectivity of a finite-list encoding; in particular $p(0, 0) = 0$ is allowed.

Stack combinators and their exact scopes. Define

$$\begin{aligned}\mathsf{Push}_m &= \mathsf{AX}_{0,0,0}; S^m; \mathsf{MP}_0,\\ \mathsf{Pop}_m &= \mathsf{AX}_{0,0,0}; S^{m+2},\\ \mathsf{Discard}_m &= \mathsf{MP}_1; \mathsf{AX}_{1,0,0}; S^{m+1}; \mathsf{MP}_0.\end{aligned}$$

Their effects are

$$\mathsf{Push}_m\langle t, w, d, h\rangle = \langle 0, [t] ::_C w, 0, 0\rangle,$$

$$\mathsf{Pop}_m\langle t, [a] ::_C w, d, h\rangle = \begin{cases}\mathsf{halt}, & t = 1,\\ \langle t, w, 0, a\rangle, & t \neq 1,\end{cases}$$

$$\mathsf{Discard}_m\langle t, w, 0, 0\rangle = \langle 0, w, 0, 0\rangle.$$

For Push, the comparison after S^m is between an implication, residue zero, and the cached universal code, residue two, so MP takes its failure branch. For Pop, precisely its final Shift tests the saved top. For Discard, the first comparison fails because $I(t, 1) > 0$, and the final comparison fails because $I(U(1, 0), 0) > 0$. Thus Pop is conditional on the saved top, while Push is total on every store and Discard is total on the displayed cache/offset interface. The compiler below permits an early halt from a saved derived one and never assumes a stronger Pop contract.

Put Offset_0 equal to the empty word and $\mathsf{Offset}_i = \mathsf{MP}_i$ for $i > 0$. From $\langle 0, w, 0, 0\rangle$ this returns $\langle 0, w, i, 0\rangle$. Then

$$\mathsf{Gen}_{m,i} = \mathsf{Push}_m; \mathsf{Offset}_i; S; \mathsf{AX}_{i,0,0}; S^{m+1}$$

returns $\langle U(i, a), w, 0, 0\rangle$ from $\langle a, w, 0, 0\rangle$, including $a = 1$.

Axiom selectors. For $1 \le j \le m - 2$, the word

$$\mathsf{Discard}_m; \mathsf{AX}_{c,D,H}; S^{j+1}; \mathsf{Pop}_m^{m-j}$$

selects the jth generated primitive head and returns it as the top, preserving the arbitrary underlying stack and clearing both caches. After S^{j+1} the top is q_j, the cache is q_{j+1}, and the remaining explicit stack is $[q_{j+2}, \ldots, q_m, 0]$. The $m - j$ Pops consume exactly this list and finish with cache zero. They cannot halt because $q_j \neq 1$. This includes the new unconditional $\Gamma(c)$ head when $m = 9$.

For $u \le a$, the restricted propositional selector is

$$\mathsf{Discard}_m; \mathsf{Offset}_u; \mathsf{AX}_{b,a-u,c}; S^m; \mathsf{Pop}_m.$$

It returns $\rho_8(a, b, c, u)$. The guarded gate selector is

$$\mathsf{Discard}_m; \mathsf{AX}_{c,b,a}; \mathsf{Pop}_m^{m+1},$$

which returns $G_m(a, b, c)$. The gate has residue zero even when the guard fails, so the Pops remain nonhalting. Soundness of the selected formulas is the retained diagonal or separate-head calculus argument; this word calculation does not replace that argument.

Modus ponens and proof compilation. The four-command suffix is unchanged:

$$\mathsf{MPSuffix}_b = S; S; \mathsf{MP}_b; S.$$

On $\langle 0, [a, I(a, b), 0] ::_C w, 0, 0\rangle$ it halts if $a = 1$ or $b = 1$; otherwise it returns $\langle b, w, 0, 0\rangle$. The dummy zero is essential to preserve the underlying stack. This computation remains valid at $a = b = 0$, where $I(0, 0) = 0$.

Recursively compile a calculus proof p into a word W_p. At an axiom use its selector. At generalization append $\mathsf{Gen}_{m,i}$. For an MP proof with minor proof p_0 of a and major proof p_1 of $I(a,b)$, use

$$W_p = \mathsf{Discard}_m; \mathsf{Push}_m; W_{p_1}; \mathsf{Push}_m; W_{p_0}; \mathsf{Push}_m; \mathsf{MPSuffix}_b.$$

The first Push reserves the dummy zero. Induction on the proof tree proves that, from every $\langle t, w, 0, 0\rangle$, W_p either halts or returns $\langle \mathrm{concl}(p), w, 0, 0\rangle$. The proof uses the exact Pop contract above, so an intermediate derived one is handled as an early halt rather than silently excluded.

For a proof of one,

$$\mathsf{AX}_{0,0,0}; W_p; \mathsf{Pop}_m$$

halts from every incoming store. The first AX clears arbitrary incoming caches, the compiled proof preserves the arbitrary residual stack, and the final Pop halts before examining that stack. This is the universal halting-word form used by the unchanged exhaustive scheduler.

In the opposite direction, annotate the live top, cache, and decoded stack entries by calculus proofs. AX introduces the retained axioms, zero, and the generalization of its old cache; successful MP records a derivation of its conclusion in the cache; failed MP returns zero; Shift moves proved formulas and halts only when it exposes one. The unconditional extra Γ is a valid primitive head in the separate calculus even when the final gate guard fails. Thus the annotation argument is unchanged apart from one additional certified stack entry.

The word construction and annotation argument are primitive recursive and use ordinary induction on finite proofs and runs. Composed with the retained calculus-to-PA translations, scheduler proof, exact startup, and positive complete physical instruction simulations, they provide the ordinary mathematical calibration for the cross-grafts. They do not assert a newly checked Lean endpoint. The independent finite-word checker verifies 64 symbolic identities for $m = 8$ and 67 for $m = 9$, with arbitrary opaque stack tails; unknown arithmetic comparisons are errors rather than guessed branches.

The zero-jump variants. The two additional variants change source site 26 from a jump to 34 into $\mathrm{dec}(R_6, 34, 27)$. Independent closure computations for both the original and changed sources give identical inductive sets of 265,964 and 263,244 source-location/zero-pattern pairs, respectively. Every pattern at site 26 has $R_6 = 0$. The two instructions therefore have identical concrete effects whenever reached from the common clean seed. Induction on source steps transports the command compiler unchanged. This is a source-invariant argument; the separate exact physical checks bind the changed row and the 58,168-step blank initialization to that source.

C Complete table of the 3,649-site construction

This is `machines/separate_combined/machine.tm`. Its start state is 0 on a blank binary tape. States $0, \ldots, 195$ are working controls; 196 is terminal. Each entry is the write symbol, one-cell direction, and next control for the indicated read symbol. The file digest is the second digest listed in Section 3.4.

Table 2: Separate-diagonal AX plus Combined exterior: complete binary table.

q	Read 0: (w, D, q')	Read 1: (w, D, q')
0	(1, R, 1)	(1, R, 61)
1	(1, R, 82)	(1, R, 1)
2	(0, L, 2)	(1, L, 3)
3	(0, L, 4)	(1, L, 3)
4	(0, L, 6)	(1, L, 3)
5	(1, R, 7)	(0, L, 5)
6	(0, L, 9)	(1, L, 3)
7	(0, R, 7)	(0, R, 10)
8	(0, R, 14)	(1, R, 13)
9	(0, L, 15)	(1, L, 3)
10	(0, L, 16)	(0, R, 10)
11	(0, L, 26)	(1, R, 18)
12	(0, R, 19)	(1, R, 20)
13	(0, L, 79)	(1, L, 31)
14	(0, L, 11)	(0, R, 22)
15	(0, L, 23)	(1, L, 3)
16	(1, L, 2)	(0, L, 2)
17	(0, R, 24)	(0, R, 32)
18	(0, R, 84)	(0, L, 95)
19	(0, R, 28)	(1, R, 29)
20	(0, R, 30)	(1, R, 31)
21	(0, R, 24)	(0, R, 57)
22	(0, R, 33)	(1, R, 22)
23	(0, L, 34)	(1, L, 3)
24	(0, L, 35)	(0, R, 1)
25	(0, R, 36)	(1, R, 25)
26	(0, R, 122)	(1, R, 41)
27	(0, R, 38)	(0, L, 39)
28	(0, R, 40)	(1, R, 41)
29	(0, R, 42)	(1, R, 43)
30	(0, R, 44)	(1, R, 45)
31	(0, R, 21)	(1, R, 46)
32	(0, R, 47)	(1, R, 32)
33	(0, R, 32)	(1, R, 33)
34	(0, L, 69)	(1, L, 3)
35	(1, R, 35)	(1, L, 2)
36	(0, R, 49)	(1, R, 36)
37	(0, R, 11)	(1, R, 101)
38	(0, L, 11)	(0, R, 36)
39	(0, L, 5)	(1, L, 51)
40	(0, R, 24)	(0, R, 25)
41	(0, R, 13)	(0, L, 16)
42	(0, R, 53)	(1, R, 54)
43	(0, R, 55)	(1, R, 56)

q	Read 0	Read 1
44	(0, R, 24)	(0, R, 52)
45	(0, R, 58)	(1, R, 59)
46	(0, R, 60)	(0, L, 16)
47	(0, R, 25)	(1, R, 47)
48	(0, L, 35)	(0, R, 61)
49	(0, R, 16)	(1, R, 49)
50	(0, R, 63)	(1, R, 64)
51	(0, L, 39)	(1, L, 51)
52	(0, R, 22)	(1, R, 52)
53	(0, L, 11)	(0, R, 49)
54	(0, R, 58)	(0, L, 16)
55	(0, R, 65)	(1, R, 66)
56	(0, R, 59)	(1, R, 67)
57	(0, R, 52)	(1, R, 57)
58	(0, R, 48)	(0, R, 25)
59	(0, L, 28)	(1, L, 68)
60	(0, L, 102)	(1, L, 30)
61	(0, R, 0)	(1, R, 61)
62	(1, R, 125)	(1, L, 51)
63	(0, R, 70)	(1, R, 196)
64	(0, L, 126)	(1, R, 31)
65	(0, R, 40)	(1, R, 60)
66	(0, L, 79)	(1, R, 71)
67	(0, R, 53)	(1, L, 72)
68	(0, R, 73)	(1, R, 74)
69	(0, R, 69)	(1, R, 75)
70	(0, R, 76)	(1, R, 77)
71	(0, R, 79)	(1, L, 110)
72	(0, L, 98)	(1, R, 54)
73	(0, R, 48)	(0, R, 33)
74	(0, R, 81)	(0, L, 41)
75	(0, R, 50)	(1, R, 132)
76	(0, R, 130)	(1, R, 21)
77	(0, R, 101)	(1, R, 90)
78	(0, R, 92)	(1, R, 93)
79	(0, R, 73)	(1, R, 72)
80	(0, R, 91)	(1, R, 119)
81	(0, R, 95)	(1, R, 96)
82	(0, R, 62)	(0, R, 1)
83	(0, R, 27)	(1, R, 97)
84	(0, R, 89)	(1, R, 88)
85	(1, L, 99)	(1, L, 85)
86	(0, R, 19)	(1, R, 100)
87	(0, R, 58)	(1, R, 58)
88	(0, R, 117)	(1, R, 115)
89	(0, L, 26)	(1, R, 134)
90	(0, R, 95)	(0, L, 131)
91	(0, R, 102)	(1, R, 103)
92	(0, R, 67)	(1, R, 56)
93	(0, R, 55)	(1, R, 104)
94	(0, R, 21)	(1, R, 105)
95	(0, R, 21)	(0, L, 10)
96	(0, R, 59)	(1, R, 41)

q	Read 0	Read 1
97	(0, R, 106)	(1, R, 8)
98	(0, R, 91)	(1, R, 107)
99	(0, L, 131)	(0, L, 85)
100	(0, R, 108)	(1, R, 94)
101	(0, L, 109)	(0, L, 2)
102	(0, R, 38)	(1, R, 68)
103	(0, R, 59)	(1, R, 110)
104	(0, R, 59)	(1, R, 111)
105	(0, R, 66)	(0, L, 16)
106	(0, R, 14)	(1, R, 110)
107	(0, R, 21)	(1, R, 112)
108	(0, R, 113)	(1, L, 20)
109	(0, R, 30)	(1, R, 65)
110	(0, R, 38)	(1, L, 108)
111	(0, R, 121)	(0, L, 16)
112	(0, R, 114)	(0, L, 16)
113	(0, R, 24)	(0, R, 47)
114	(0, R, 48)	(0, R, 32)
115	(0, R, 70)	(1, R, 108)
116	(0, R, 19)	(1, R, 118)
117	(0, R, 126)	(1, R, 28)
118	(0, R, 122)	(1, R, 107)
119	(0, R, 94)	(1, R, 86)
120	(0, R, 83)	(1, R, 98)
121	(0, R, 87)	(1, R, 87)
122	(0, R, 17)	(1, L, 109)
123	(0, R, 70)	(1, R, 101)
124	(0, R, 96)	(1, R, 108)
125	(1, L, 39)	(1, L, 99)
126	(0, R, 80)	(1, R, 78)
127	(0, R, 133)	(1, R, 20)
128	(0, R, 124)	(1, R, 123)
129	(0, R, 123)	(1, R, 128)
130	(0, R, 44)	(1, R, 90)
131	(0, L, 5)	(0, L, 131)
132	(0, R, 127)	(1, R, 59)
133	(0, R, 120)	(1, R, 37)
134	(0, R, 135)	(1, R, 186)
135	(0, R, 136)	(1, R, 140)
136	(0, R, 12)	(1, R, 137)
137	(0, R, 86)	(1, R, 138)
138	(0, R, 139)	(1, R, 58)
139	(0, R, 78)	(1, R, 58)
140	(0, R, 141)	(1, R, 176)
141	(0, R, 142)	(1, R, 143)
142	(0, R, 64)	(1, R, 12)
143	(0, R, 144)	(1, R, 175)
144	(0, R, 86)	(1, R, 145)
145	(0, R, 146)	(1, R, 170)
146	(0, R, 147)	(1, R, 162)
147	(0, R, 78)	(1, R, 148)
148	(0, R, 149)	(1, R, 161)
149	(0, R, 126)	(1, R, 150)

q	Read 0	Read 1
150	(0, R, 151)	(1, R, 156)
151	(0, R, 156)	(1, R, 153)
152	(0, L, 146)	(1, R, 78)
153	(0, R, 154)	(1, R, 155)
154	(0, R, 116)	(1, R, 152)
155	(0, R, 152)	(1, R, 136)
156	(0, L, 143)	(1, R, 152)
157	(0, R, 158)	(1, R, 159)
158	(0, R, 159)	(1, R, 137)
159	(0, R, 19)	(1, R, 160)
160	(0, R, 58)	(1, L, 192)
161	(0, R, 162)	(1, R, 164)
162	(0, R, 12)	(1, R, 163)
163	(0, R, 116)	(1, R, 138)
164	(0, L, 179)	(1, R, 166)
165	(0, R, 137)	(1, R, 136)
166	(0, R, 167)	(1, R, 168)
167	(0, R, 163)	(1, R, 78)
168	(0, L, 146)	(1, L, 175)
169	(0, R, 86)	(1, R, 139)
170	(0, R, 171)	(1, R, 172)
171	(0, R, 172)	(1, R, 154)
172	(0, R, 173)	(1, R, 58)
173	(0, R, 160)	(1, R, 155)
174	(0, R, 116)	(1, R, 169)
175	(0, R, 157)	(1, R, 174)
176	(0, R, 177)	(1, R, 184)
177	(0, R, 178)	(1, R, 166)
178	(0, R, 58)	(1, R, 179)
179	(0, R, 154)	(1, R, 180)
180	(0, R, 165)	(1, R, 181)
181	(0, R, 152)	(1, R, 182)
182	(0, R, 116)	(1, R, 183)
183	(0, R, 164)	(1, R, 168)
184	(0, R, 185)	(1, R, 185)
185	(0, L, 153)	(1, R, 174)
186	(0, R, 187)	(1, R, 195)
187	(0, R, 188)	(1, R, 193)
188	(0, R, 116)	(1, R, 189)
189	(0, R, 190)	(1, R, 191)
190	(0, R, 168)	(1, R, 78)
191	(0, R, 158)	(1, R, 192)
192	(0, R, 185)	(1, R, 190)
193	(0, R, 167)	(1, R, 194)
194	(0, R, 159)	(1, R, 122)
195	(0, R, 129)	(0, L, 131)